

Assignment: Preparing Data for Analysis

Objective:

This assignment aims to help you develop essential data cleaning skills using Power Query. You will be working with a dataset that contains multiple tables, and your task is to identify data issues such as missing values, duplicates, inconsistencies, and incorrect formats. After identifying these problems, you will clean the data and prepare it for analysis.

You will also explore how to transform and organize data in Power BI's Power Query to make it ready for reporting and visualisation.

Dataset:  Customers ,  Stocks ,  Transaction_1 ,  Transaction_2

Dataset Overview:

You will be provided with the following tables, each containing relevant data for analysis:

- **Customers**
This table contains information about the customers, including their unique identifier and other personal details.
- **Stock**
This table includes stock data for different products over time.
- **Transaction_1 and Transaction_2**
These tables contain transaction details for sales made by customers, including product purchases and associated information.

Instructions:

You are required to load the dataset into Excel. Once you have the data loaded, thoroughly explore the tables and identify any data quality issues. You are expected to clean and transform the data in Power Query based on your findings.

Here are some general guidelines for the task:

- **Explore the Data:** Understand the structure of the tables and relationships between them. Look for any missing or incorrect values, duplicates, or other anomalies that could affect your analysis.
- **Clean the Data:** Identify data quality issues and address them appropriately. Pay attention to formatting errors, missing data, and any inconsistencies that may hinder analysis.
- **Create Transformations:** You may need to create new calculated columns or measures to ensure the data is ready for analysis and reporting. For example, transforming stock data or calculating total sales for each customer.

Deliverables:

- An Excel file (.xlsx) with the cleaned data.
- A summary (200-300 words) explaining the issues you encountered, the steps you took to clean the data, and how you prepared it for analysis.